Economía Laboral — Trabajo Práctico N.º 2

Parte I

Titular: Julio J. Elías Ayudante: Juan M. Favetto

1. **Problemas del Capítulo 2 de Borjas, *Labor Economics*.**
   (a) Currently a firm pays 10 % of each employee's salary into a retirement account, regardless of whether the employee also contributes to the account. The firm is considering changing this system to a 10 % match, meaning that the firm will match, up to 10 % of each employee's salary, the employee's contributions into the account. Some people at the firm think this change will lead employees to save more and therefore to be able to afford to retire, while others at the firm believe this will lead employees to have less retirement savings and therefore be less able to afford to retire. Explain why either point of view could be correct.

2. **Oferta de trabajo. Modelo estático.** Considere el siguiente modelo estático de oferta de trabajo. Un individuo vive por un período y tiene la siguiente función de utilidad

$$u(c, l) = c^{1-\alpha}\, l^{\alpha},$$

en donde c es consumo, l horas de ocio y T es la dotación de horas de tiempo que posee disponibles el individuo.
   (a) Escriba la restricción presupuestaria del individuo asumiendo que el individuo posee ingresos no salariales A.
   (b) Escriba el Lagrangiano de este problema.
   (c) Calcule las condiciones de primer orden.
   (d) Derive la oferta de trabajo del individuo.
   (e) Calcule la elasticidad de la oferta de trabajo. ¿De qué variables depende?
   (f) Calcule el salario de reserva. ¿De qué variables depende?
   (g) Analice el efecto de un impuesto al ingreso laboral sobre la oferta de trabajo, el salario de reserva y el consumo.
   (h) Analice el efecto de un impuesto sobre el consumo sobre la oferta de trabajo, el salario de reserva y el consumo. Compare con (g).
   (i) Analice el efecto de un impuesto al ingreso no laboral sobre la oferta de trabajo, el salario de reserva y el consumo. Compare con (g) y (h).

   Asuma que existen 1.000 individuos idénticos en la economía.
   (j) Calcule la oferta de trabajo agregada.
   (k) Calcule la elasticidad de la oferta de trabajo agregada.

   Ahora asuma que la demanda agregada de trabajo por parte de las firmas es inelástica e igual a 1.000 horas.
   (l) Calcule el salario de equilibrio, el nivel de consumo y las horas de trabajo individuales.
   (m) Calcule el salario de equilibrio, el nivel de consumo y las horas de trabajo individuales asumiendo que se impone un impuesto al ingreso laboral.

(n) ¿Quién paga el impuesto? ¿Las firmas o los trabajadores?

(ñ) Calcule el salario de equilibrio, el nivel de consumo y las horas de trabajo individuales asumiendo que se impone un impuesto al ingreso no laboral.

Ahora asuma que hay heterogeneidad entre los individuos de acuerdo al ingreso no laboral que poseen. 500 individuos poseen un ingreso no laboral A_l y 500 un ingreso no laboral $A_h > A_l$.

(o) Calcule la elasticidad de la oferta de trabajo. ¿De qué variables depende? Compare con (k).

(p) Calcule el salario de equilibrio, las horas trabajadas y los niveles de consumo para ambos grupos. ¿Quiénes trabajan más? ¿Quiénes están mejor en términos del nivel de bienestar?

3. **Racionalidad.** ¿Considera racionales los siguientes comportamientos? Explique.
   (a) Sherwin Rosen stores a young bottle of wine that cost a few bucks. By luck the bottle turns out to be worth several hundred dollars after ten years. But Sherwin Rosen refuses to sell, even though he would never contemplate paying that much for an otherwise equivalent bottle.
   (b) A family may refuse to sell for half-a-million dollars the house it has lived in for twenty years, even though it would be unwilling to spend anywhere near that amount for an otherwise equivalent house.
   (c) Mr. H mows his own lawn. His neighbor's son would mow it for $8. He wouldn't mow his neighbor's same-sized lawn for $20.

4. **Shock tecnológico.** Consider a competitive firm that experiences a positive technology shock that increases its productivity: at every level of employment, the firm can now produce 200 additional units of output compared to before.
   (a) How does this technology shock affect the firm's optimal level of employment?
   (b) What is the effect on the firm's marginal cost?
   (c) What happens to the market price of the good produced and sold by the firm?

5. **Firma e industria.** An industry has a (large) fixed number of identical firms. Each firm produces the single output Y, with labor and capital according to the production function $Y = F(L, K)$. The industry as a whole faces a downward-sloping demand for its output and an upward-sloping supply of labor. Capital is perfectly elastically supplied to the industry and all markets are competitive.
   (a) How will the elasticity of supply for an individual firm compare to the elasticity of supply for the industry as a whole?
   (b) How will the elasticity of demand for labor for an individual firm compare to the elasticity of demand for labor for the industry as a whole?
   (c) How will the effect of a change in the price of capital on output differ for an individual firm and the industry as a whole?
   (d) How will the effect of a change in the price of capital on the quantity of labor utilized differ for an individual firm and the industry as a whole?

6. **Mecanización e inteligencia artificial.** Compare the effects of mechanization during the Industrial Revolution and artificial intelligence (AI) today on labor demand.
   (a) How did the introduction of mechanization affect labor demand across different tasks and skill levels during the Industrial Revolution? In your answer, discuss the distinction

between labor-saving and labor-augmenting technological change.

(b) How does AI compare to mechanization in terms of its impact on labor demand today? Which types of tasks or skills are most likely to be complemented by AI, and which are most likely to be substituted?

(c) Drawing on Joel Mokyr's view that major technological revolutions tend to create new tasks and occupations rather than simply eliminate jobs, discuss whether AI is likely to reduce total labor demand in the long run or mainly reshape the composition of employment.

(d) What role do institutions, education, and adaptability of the workforce play in determining whether a technological revolution leads to sustained employment growth?